Jeremy Ku-Benjet

Contact Information

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Academic Interests

Applied Programming Languages, Compilers, ASIC Design, EDA Tooling, Hardware Description Languages, Software Engineering

Education

Cornell University, Ithaca, New York USA

M.S., Computer Science, August 2025 (expected graduation date: May 2027)

- Advisor: Adrian Sampson

B.S., Computer Science, May 2025

Stuyvesant High School, New York, New York USA (June 2022)

Academic Experience

Cornell University, Ithaca, New York USA

Researcher on fud2: An Extensible Compiler Driver **Summer 2024 - Fall 2025**

fud2 is an extensible compiler driver for hardware tooling. It is the first compiler driver framework we are aware of that provides a declarative way to integrate new backend tools. fud2 is the primary way to compile and run code in the Calyx ecosystem. Advised by Adrian Sampson.

- Extended Rhai scripting language to act as a configuration language for fud2.
- Wrote an algorithm to automatically synthesize tool flows from simple CLI commands.
- Presented at LATTE'26, a workshop about hardware acceleration and programming languages co-located with ASPLOS.

Backend Developer on CMSX **Spring 2024 - Present**

CMSX is a course management solution started at Cornell University in 2002. It is still actively developed and used by thousands of students over a variety of coureses.

- Lead design and implementation of a rubrics feature and implemented backend grading logic
- Audited a transaction based database system for errors when rolling back invalid database queries
- Add a calendar feature allowing students to download deadlines using ical calendar format

Engineer on CGRA Tapeout **Spring 2026 - Present**

Worked with another student to tape out a CGRA using the TSMC 180nm PDK. Worked across the stack on design, RTL implementation, verification, and physical design. Currently awaiting packaging to start testing.

Teaching Assistant **January 2023 - Present**

Taught lab sections, designed assignments and infrastructure, wrote exam questions, and graded for a variety of courses in systems, software engineering, computer architecture, programming languages, and compilers.

Researcher on As Yet Unnamed Project **Summer 2026 - Present**

Thesis work on an early stage project creating a novel macro interface and tooling to simplify engineering effort in physical design and speed up PnR. Advised by Adrian Sampson.

Computer Skills

Languages: Rust, Java, C, C++, Python, OCaml, Verilog/SystemVerilog, RISC-V and x86-64 assembly, Bash, TCL.

Tools/Technologies: Git, LLVM, Unix Shell Scripting, Docker/Podman, Kubernetes, Librelane, SDC.